A REFINED VALUATIVE CRITERION WITH A DENSE SUBSCHEME IMPLIES UNIVERSAL CLOSEDNESS, AND SEPARATEDNESS WHEN THE MORPHISM IS QUASI-SEPARATED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-refined-valuative-criterion-universally-closed is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 10219-10234 and proof span 10236-10327 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The statement and proof contain four xymatrix solid diagrams with a dotted arrow; they are copied verbatim and the retained chapter preamble loads Xy-pic.

Editorial note. The proof uses the dominance relation on valuation rings inside a field exactly as the author writes it, including the conclusion that the given valuation ring equals the intersection of the field with the ambient residue field.

Editorial note. 7 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The characterization of universally closed morphisms by the existence part of the valuative criterion, cited from the schemes chapter. Cited at proof lines 10237-10239. Reduces universal closedness of the given morphism to the existence of the dotted arrow in a valuation ring diagram.
- Reductions of schemes and the behaviour of morphisms and dense images under passage to the reduction, together with the description of the scheme theoretic image in the reduced case, cited from the schemes chapter and the same chapter. Cited at proof lines 10244-10249. Lets the proof assume all schemes reduced and rewrite density of the image as the scheme theoretic image being the whole space.
- Flatness of a polynomial algebra over the coordinate ring of an affine open and invariance of the scheme theoretic image under flat base change, cited from the same chapter. Cited at proof lines 10256-10259. Shows that the

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scheme theoretic image of the base changed dense open is the whole of the polynomial algebra's spectrum.
- Existence of a valuation ring diagram reaching a prescribed point of a scheme theoretic image, cited from the same chapter. Cited at proof lines 10260-10262. Produces the valuation ring A' and the field K' whose closed point maps to the point of interest.
- Existence of a valuation ring inside a residue field dominating the image of a given valuation ring, and the description of a valuation ring sandwiched between a valuation ring and a residue field, both cited from the algebra chapter. Cited at proof lines 10271-10278. Builds the valuation ring A" and the auxiliary valuation ring C that put the situation into the shape of the hypothesis.

Source reference. Original lemma 00IA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 672–672. Same-chapter reference used in the retained author text at source line(s) 941, 10279. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 088Z: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10479–10479. Same-chapter reference used in the retained author text at source line(s) 10284. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 081I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10218–10218. Same-chapter reference used in the retained author text at source line(s) 10264. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 02JQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3345–3345. Same-chapter reference used in the retained author text at source line(s) 1367, 4438, 10265, 14326. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

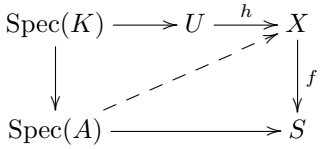
Source reference. Original lemma 056B: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6704–6704. Same-chapter reference used in the retained author text at source line(s) 1207, 10253. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0356: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4117–4117. Same-chapter reference used in the retained author text at source line(s) 291, 987, 10250, 13640, 14408, 14563. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original proposition 01KF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2048–2048. Same-chapter reference used

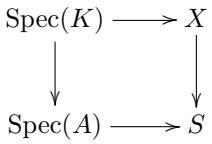
in the retained author text at source line(s) 10208, 10239, 10613. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $f : X \to S$ and $h : U \to X$ be morphisms of schemes. Assume that f and h are quasi-compact and that $h(U)$ is dense in X. If given any commutative solid diagram*



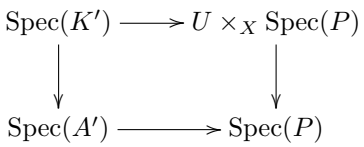
where A is a valuation ring with field of fractions K, there exists a unique dotted arrow making the diagram commute, then f is universally closed. If moreover f is quasi-separated, then f is separated.

Proof. To prove f is universally closed we will verify the existence part of the valuative criterion for f which suffices by Schemes, Proposition 01KF. To do this, consider a commutative diagram



where A is a valuation ring and K is the fraction field of A. Note that since valuation rings and fields are reduced, we may replace U, X, and S by their respective reductions by Schemes, Lemma 0356. In this case the assumption that $h(U)$ is dense means that the scheme theoretic image of $h : U \to X$ is X, see Lemma 056B. We may also replace S by an affine open through which the morphism $\mathrm{Spec}(A) \to S$ factors. Thus we may assume that $S = \mathrm{Spec}(R)$.

Let $\mathrm{Spec}(B) \subset X$ be an affine open through which the morphism $\mathrm{Spec}(K) \to X$ factors. Choose a polynomial algebra P over B and a B-algebra surjection $P \to K$. Then $\mathrm{Spec}(P) \to X$ is flat. Hence the scheme theoretic image of the morphism $U \times_X \mathrm{Spec}(P) \to \mathrm{Spec}(P)$ is $\mathrm{Spec}(P)$ by Lemma 081I. By Lemma 02JQ we can find a commutative diagram



where A' is a valuation ring and K' is the fraction field of A' such that the closed point of $\mathrm{Spec}(A')$ maps to $\mathrm{Spec}(K) \subset \mathrm{Spec}(P)$. In other words, there is a B-algebra map $\varphi : K \to A'/\mathfrak{m}_{A'}$. Choose a valuation ring $A'' \subset A'/\mathfrak{m}_{A'}$ dominating $\varphi(A)$ with field of fractions $K'' = A'/\mathfrak{m}_{A'}$ (Algebra, Lemma 00IA). We set

$$C = \{\lambda \in A' \mid \lambda \bmod \mathfrak{m}_{A'} \in A''\}.$$

which is a valuation ring by Algebra, Lemma 088Z. As C is an R-algebra with fraction field K', we obtain a commutative diagram

$$\begin{array}{ccccc} \mathrm{Spec}(K') & \longrightarrow & U & \longrightarrow & X \\ \downarrow & & & \nearrow & \downarrow \\ \mathrm{Spec}(C) & & \longrightarrow & & S \end{array}$$

as in the statement of the lemma. Thus a dotted arrow fitting into the diagram as indicated. By the uniqueness assumption of the lemma the composition $\mathrm{Spec}(A') \to \mathrm{Spec}(C) \to X$ agrees with the given morphism $\mathrm{Spec}(A') \to \mathrm{Spec}(P) \to \mathrm{Spec}(B) \subset X$. Hence the restriction of the morphism to the spectrum of $C/\mathfrak{m}_{A'} = A''$ induces the given morphism $\mathrm{Spec}(K'') = \mathrm{Spec}(A'/\mathfrak{m}_{A'}) \to \mathrm{Spec}(K) \to X$. Let $x \in X$ be the image of the closed point of $\mathrm{Spec}(A'') \to X$. The image of the induced ring map $\mathcal{O}_{X,x} \to A''$ is a local subring which is contained in $K \subset K''$. Since A is maximal for the relation of domination in K and since $A \subset A''$, we have $A = K \cap A''$. We conclude that $\mathcal{O}_{X,x} \to A''$ factors through $A \subset A''$. In this way we obtain our desired arrow $\mathrm{Spec}(A) \to X$.

Finally, assume f is quasi-separated. Then $\Delta : X \to X \times_S X$ is quasi-compact. Given a solid diagram

$$\begin{array}{ccccc} \mathrm{Spec}(K) & \longrightarrow & U & \xrightarrow{h} & X \\ \downarrow & & & \nearrow & \downarrow \Delta \\ \mathrm{Spec}(A) & & \longrightarrow & & X \times_S X \end{array}$$

where A is a valuation ring with field of fractions K, there exists a unique dotted arrow making the diagram commute. Namely, the lower horizontal arrow is the same thing as a pair of morphisms $\mathrm{Spec}(A) \to X$ which can serve as the dotted arrow in the diagram of the lemma. Thus the required uniqueness shows that the lower horizontal arrow factors through Δ. Hence we can apply the result we just proved to $\Delta : X \to X \times_S X$ and $h : U \to X$ and conclude that Δ is universally closed. Clearly this means that f is separated. $\square$